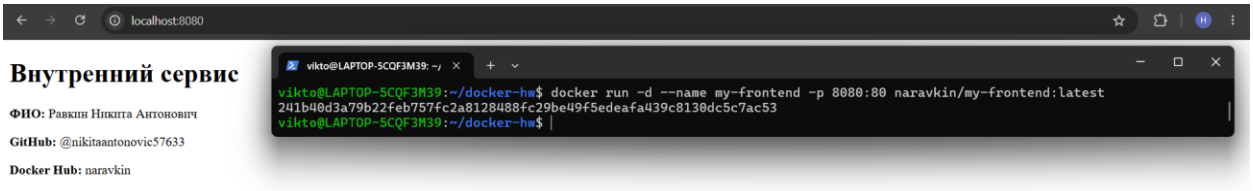
Лабораторная работа №2. Docker

1.1. docker build -t naravkin/my-frontend:latest .

```
vikto@LAPTOP-5CQF3M39:~/docker-hw$ docker build -t naravkin/my-frontend:latest .
[+] Building 68.1s (8/8) FINISHED                                                                                docker:default
 => [internal] load build definition from Dockerfile                                                                      0.0s
 => => transferring dockerfile: 104B                                                                                      0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                                                          10.1s
 => [auth] library/nginx:pull token for registry-1.docker.io                                                              0.0s
 => [internal] load .dockerignore                                                                                         0.0s
 => => transferring context: 2B                                                                                           0.0s
 => [internal] load build context                                                                                         0.0s
 => => transferring context: 500B                                                                                         0.0s
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:a3554c0d0b48177b1413c26e01f695334428020d9df01243fbae3e11cfd          56.6s
 => => resolve docker.io/library/nginx:alpine@sha256:a3554c0d0b48177b1413c26e01f695334428020d9df01243fbae3e11cfd1          0.0s
 => => sha256:b9b388a21ecfde6ac80505f23f527146a2129dfb104e79b029bbd00d4cc1a919 20.53MB / 20.53MB                         56.0s
 => => sha256:9b0f7dfe35b2711a17f60dadd6d4035dc0cdee152538cc5c0987029e94679548 1.40kB / 1.40kB                            0.7s
 => => sha256:26185d08a3ec318dd85a6c9f2172e714ec13cececd7f33c533b40b1689f9fedb 404B / 404B                                0.6s
 => => sha256:2ec5fb117ffa9a78b15fb6a78745ffc569c8da0aee9f36ace4cd67a44b1fef79 1.21kB / 1.21kB                            0.9s
 => => sha256:7cc330898115ce55acccb0cc1e7dd091df4410c8c5f3c540710ee0ac689f5fcb 964B / 964B                                0.2s
 => => sha256:657023555a2a19dc6036468198bb350d2c7feaac7f2d0a3901b4dd3cb25e40c5 629B / 629B                                0.2s
 => => sha256:12b85cff06fd7d383d36a9c96503fe049c9d4bf3dfda5cbb9ffa6ae99b9c07a6 1.94MB / 1.94MB                            3.9s
 => => sha256:7b229517ed222597277036d94892173207d01a731ab3a88eeb1de2e963a2c788 3.85MB / 3.85MB                            6.5s
 => => extracting sha256:7b229517ed222597277036d94892173207d01a731ab3a88eeb1de2e963a2c788                                 0.2s
 => => extracting sha256:12b85cff06fd7d383d36a9c96503fe049c9d4bf3dfda5cbb9ffa6ae99b9c07a6                                 0.2s
 => => extracting sha256:657023555a2a19dc6036468198bb350d2c7feaac7f2d0a3901b4dd3cb25e40c5                                 0.0s
 => => extracting sha256:7cc330898115ce55acccb0cc1e7dd091df4410c8c5f3c540710ee0ac689f5fcb                                 0.0s
 => => extracting sha256:26185d08a3ec318dd85a6c9f2172e714ec13cececd7f33c533b40b1689f9fedb                                 0.0s
 => => extracting sha256:2ec5fb117ffa9a78b15fb6a78745ffc569c8da0aee9f36ace4cd67a44b1fef79                                 0.0s
 => => extracting sha256:9b0f7dfe35b2711a17f60dadd6d4035dc0cdee152538cc5c0987029e94679548                                 0.0s
 => => extracting sha256:b9b388a21ecfde6ac80505f23f527146a2129dfb104e79b029bbd00d4cc1a919                                 0.5s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html                                                                0.1s
 => exporting to image                                                                                                    1.1s
 => => exporting layers                                                                                                   0.1s
 => => exporting manifest sha256:42e8f8246c43fbf5fd37e6b2975df023014e1ba92d1d72a2e631774fe85f8d37                         0.0s
 => => exporting config sha256:479f0f35090ec3a25cb81ae72dc7a96ddbfae59ad67ce9b739d1d5313a946494                           0.0s
 => => exporting attestation manifest sha256:76fcd62299f0013ee477cb716073989c15b3c6e15334d8c38929ed670e4d6dda             0.0s
 => => exporting manifest list sha256:9a17c9fac46fe30e3f74a562b1d0f222491d919a1092b4e1d2d82e72cc42670c                    0.0s
 => => naming to docker.io/naravkin/my-frontend:latest                                                                    0.0s
 => => unpacking to docker.io/naravkin/my-frontend:latest                                                                 0.9s
```

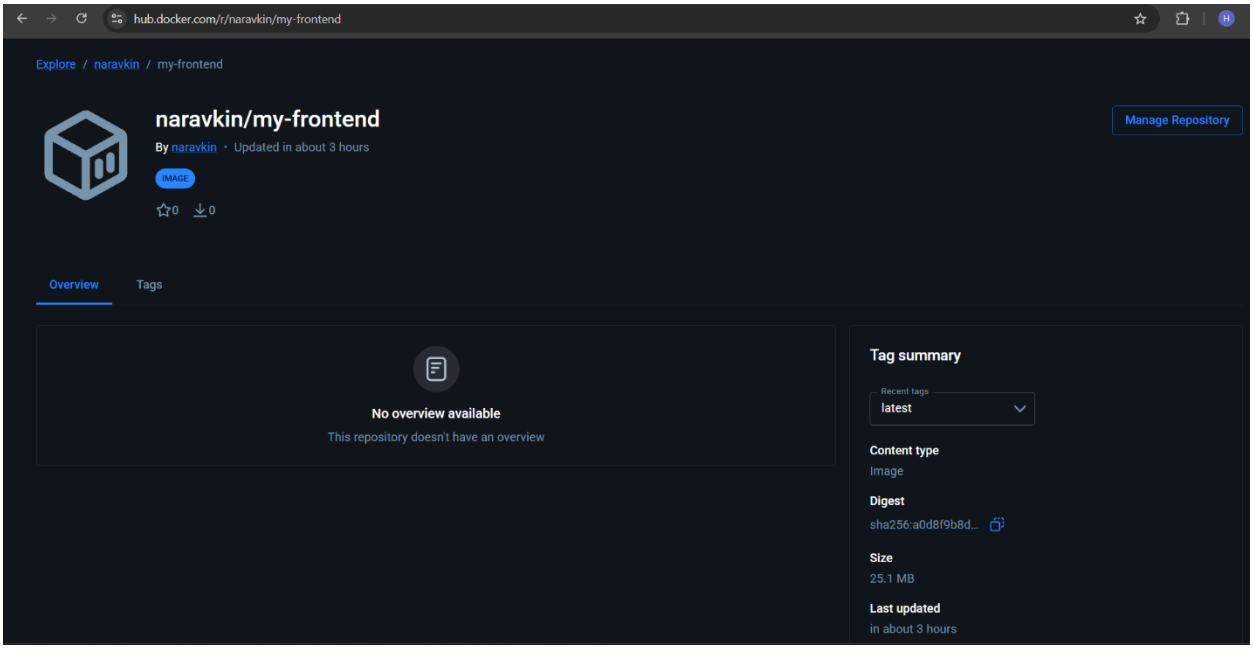
1.2. docker run -d --name my-frontend -p 8080:80 naravkin/my-frontend:latest



1.3. docker push naravkin/my-frontend:latest

```
vikto@LAPTOP-5CQF3M39:~/docker-hw$ docker push naravkin/my-frontend:latest
The push refers to repository [docker.io/naravkin/my-frontend]
44136fa355b3: Pushed
12b85cff06fd: Pushed
3e566f7357e1: Pushed
7b229517ed22: Pushed
657023555a2a: Pushed
7cc330898115: Pushed
26185d08a3ec: Pushed
2ec5fb117ffa: Pushed
9b0f7dfe35b2: Pushed
b9b388a21ecf: Pushed
46200fb2cbc5: Pushed
latest: digest: sha256:a0d8f9b8d03e223f3842e91478850038f15834dde27bfc75b7b18c52fcb24e40 size: 856
```

1.4. https://hub.docker.com/r/naravkin/my-frontend



1.5. docker rmi naravkin/my-frontend:latest
docker pull naravkin/my-frontend:latest

```
vikto@LAPTOP-5CQF3M39:~/docker-hw$ docker rmi naravkin/my-frontend:latest
Error response from daemon: conflict: unable to delete naravkin/my-frontend:latest (must be forced) - container 241b40d3a79b is using its referenced image a
0d8f9b8d03e
vikto@LAPTOP-5CQF3M39:~/docker-hw$ docker rm -f 241b40d3a79b
241b40d3a79b
vikto@LAPTOP-5CQF3M39:~/docker-hw$ docker rmi naravkin/my-frontend:latest
Untagged: naravkin/my-frontend:latest
Deleted: sha256:a0d8f9b8d03e223f3842e91478850038f15834dde27bfc75b7b18c52fcb24e40
vikto@LAPTOP-5CQF3M39:~/docker-hw$ docker pull naravkin/my-frontend:latest
latest: Pulling from naravkin/my-frontend
12b85cff06fd: Pull complete
46200fb2cbc5: Pull complete
657023555a2a: Pull complete
7cc330898115: Pull complete
26185d08a3ec: Pull complete
2ec5fb117ffa: Pull complete
7b229517ed22: Pull complete
9b0f7dfe35b2: Pull complete
b9b388a21ecf: Pull complete
44136fa355b3: Download complete
3e566f7357e1: Download complete
Digest: sha256:a0d8f9b8d03e223f3842e91478850038f15834dde27bfc75b7b18c52fcb24e40
Status: Downloaded newer image for naravkin/my-frontend:latest
docker.io/naravkin/my-frontend:latest
```

2.1.

```
vikto@LAPTOP-5CQF3M39: ~/
 GNU nano 8.7.1                                  docker-compose.yml
services:
  frontend:
    build: .
    image: naravkin/my-frontend:latest
    ports:
      - "8080:80"
    depends_on:
      api:
        condition: service_started

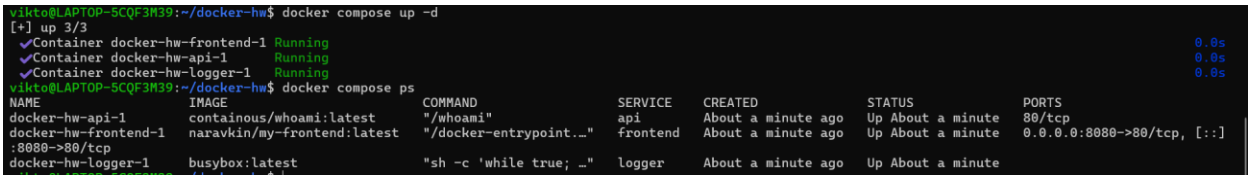
  api:
    image: containous/whoami:latest

  logger:
    image: busybox:latest
    command: >
      sh -c "while true; do echo \"$(date '+%Y-%m-%d %H:%M:%S') - log entry\" >> /data/log.txt; sleep 5; done"
    volumes:
      - logger_data:/data

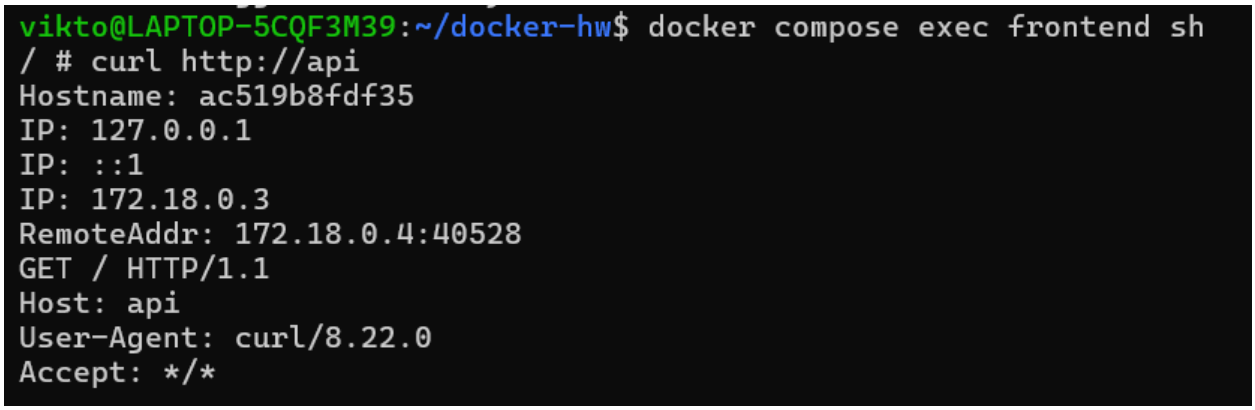
volumes:
  logger_data:
    name: logger_data

                                [ Read 23 lines ]
^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo         M-A Set Mark     M-] To Bracket
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo         M-6 Copy         ^B Where Was
```

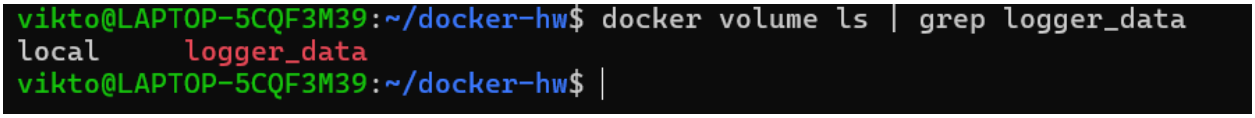
2.2. docker compose up -d
docker compose ps



2.3. docker compose exec frontend sh
curl http://api



2.4. docker volume ls | grep logger_data



2.5. docker compose exec logger cat /data/log.txt

```
vikto@LAPTOP-5CQF3M39:~/docker-hw$ docker compose exec logger cat /data/log.txt
2026-10-22 15:11:30 - log entry
2026-10-22 15:11:35 - log entry
2026-10-22 15:11:40 - log entry
2026-10-22 15:11:45 - log entry
2026-10-22 15:11:50 - log entry
2026-10-22 15:11:55 - log entry
2026-10-22 15:12:00 - log entry
2026-10-22 15:12:05 - log entry
2026-10-22 15:12:10 - log entry
2026-10-22 15:12:15 - log entry
2026-10-22 15:12:20 - log entry
2026-10-22 15:12:25 - log entry
2026-10-22 15:12:30 - log entry
2026-10-22 15:12:35 - log entry
2026-10-22 15:12:40 - log entry
2026-10-22 15:12:45 - log entry
2026-10-22 15:12:50 - log entry
2026-10-22 15:12:55 - log entry
2026-10-22 15:13:00 - log entry
2026-10-22 15:13:05 - log entry
2026-10-22 15:13:10 - log entry
2026-10-22 15:13:15 - log entry
2026-10-22 15:13:20 - log entry
2026-10-22 15:13:25 - log entry
2026-10-22 15:13:30 - log entry
2026-10-22 15:13:35 - log entry
2026-10-22 15:13:40 - log entry
2026-10-22 15:13:45 - log entry
2026-10-22 15:13:50 - log entry
2026-10-22 15:13:55 - log entry
2026-10-22 15:14:00 - log entry
2026-10-22 15:14:05 - log entry
2026-10-22 15:14:10 - log entry
```

3.1. docker compose down

docker compose up -d

```
vikto@LAPTOP-5CQF3M39:~/docker-hw$ docker compose down
[+] down 4/4
 ✔Container docker-hw-logger-1   Removed                                    3.4s
 ✔Container docker-hw-frontend-1 Removed                                    0.6s
 ✔Container docker-hw-api-1      Removed                                    0.3s
 ✔Network docker-hw_default      Removed                                    0.3s
vikto@LAPTOP-5CQF3M39:~/docker-hw$ docker compose up -d
[+] up 4/4
 ✔Network docker-hw_default      Created                                    0.1s
 ✔Container docker-hw-logger-1   Started                                    1.3s
 ✔Container docker-hw-api-1      Started                                    0.7s
 ✔Container docker-hw-frontend-1 Started                                    0.9s
```

3.2. docker compose exec logger cat /data/log.txt

```
vikto@LAPTOP-5CQF3M39:~/docker-hw$ docker compose exec logger cat /data/log.txt
2026-10-22 15:11:30 - log entry
2026-10-22 15:11:35 - log entry
2026-10-22 15:11:40 - log entry
2026-10-22 15:11:45 - log entry
2026-10-22 15:11:50 - log entry
2026-10-22 15:11:55 - log entry
2026-10-22 15:12:00 - log entry
2026-10-22 15:12:05 - log entry
2026-10-22 15:12:10 - log entry
2026-10-22 15:12:15 - log entry
2026-10-22 15:12:20 - log entry
2026-10-22 15:12:25 - log entry
2026-10-22 15:12:30 - log entry
2026-10-22 15:12:35 - log entry
2026-10-22 15:12:40 - log entry
2026-10-22 15:12:45 - log entry
2026-10-22 15:12:50 - log entry
2026-10-22 15:12:55 - log entry
2026-10-22 15:13:00 - log entry
2026-10-22 15:13:05 - log entry
2026-10-22 15:13:10 - log entry
2026-10-22 15:13:15 - log entry
2026-10-22 15:13:20 - log entry
2026-10-22 15:13:25 - log entry
2026-10-22 15:13:30 - log entry
2026-10-22 15:13:35 - log entry
2026-10-22 15:13:40 - log entry
2026-10-22 15:13:45 - log entry
2026-10-22 15:13:50 - log entry
2026-10-22 15:13:55 - log entry
2026-10-22 15:14:00 - log entry
2026-10-22 15:14:05 - log entry
2026-10-22 15:14:10 - log entry
2026-10-22 15:14:15 - log entry
2026-10-22 15:14:20 - log entry
2026-10-22 15:14:25 - log entry
2026-10-22 15:14:30 - log entry
2026-10-22 15:14:35 - log entry
2026-10-22 15:14:40 - log entry
2026-10-22 16:01:03 - log entry
2026-10-22 16:01:08 - log entry
2026-10-22 16:01:13 - log entry
2026-10-22 16:01:18 - log entry
2026-10-22 16:01:23 - log entry
2026-10-22 16:01:28 - log entry
2026-10-22 16:01:33 - log entry
2026-10-22 16:01:38 - log entry
2026-10-22 16:01:43 - log entry
2026-10-22 16:01:48 - log entry
2026-10-22 16:01:53 - log entry
2026-10-22 16:01:58 - log entry
2026-10-22 16:02:03 - log entry
2026-10-22 16:02:08 - log entry
2026-10-22 16:02:13 - log entry
2026-10-22 16:02:18 - log entry
2026-10-22 16:02:23 - log entry
2026-10-22 16:02:28 - log entry
2026-10-22 16:02:33 - log entry
2026-10-22 16:02:38 - log entry
2026-10-22 16:02:43 - log entry
2026-10-22 16:02:48 - log entry
2026-10-22 16:02:53 - log entry
2026-10-22 16:02:58 - log entry
2026-10-22 16:03:03 - log entry
2026-10-22 16:03:08 - log entry
2026-10-22 16:03:23 - log entry
2026-10-22 16:03:28 - log entry
2026-10-22 16:03:33 - log entry
2026-10-22 16:03:39 - log entry
2026-10-22 16:03:44 - log entry
2026-10-22 16:03:49 - log entry
2026-10-22 16:03:54 - log entry
2026-10-22 16:03:59 - log entry
2026-10-22 16:04:04 - log entry
2026-10-22 16:04:09 - log entry
2026-10-22 16:04:14 - log entry
2026-10-22 16:04:19 - log entry
2026-10-22 16:04:24 - log entry
2026-10-22 16:04:29 - log entry
vikto@LAPTOP-5CQF3M39:~/docker-hw$
```

3.3. docker compose down -v
docker compose up -d
docker compose exec logger sh -c "cat /data/log.txt "

```
vikto@LAPTOP-5CQF3M39:~/docker-hw$ docker compose exec logger sh -c "cat /data/log.txt"
2026-10-22 15:06:07 - log entry
2026-10-22 15:06:12 - log entry
2026-10-22 15:06:17 - log entry
```